

ROOT, REALITY, LITERAL, AND SIMULATION STRUCTURES

A Reference-Frame Study of What We Currently Understand — to Aid Later Design of Conditional Failure at the Root Level

Correction brief: the object of study is the structures. CFTRL is a downstream use of that study, not the title of the study.

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Relation to CFTRL	Upstream study. CFTRL consumes this map. This document is not a CFTRL essay with a new subtitle.
Corrects	Title and subject inversion in prior collaborator files, including Grok RES-002, LAB-003, and OUT-004

1. Acknowledgment of the discrepancy

The Architect is right. The collaboration treated “Conditional Failure at the Root Level” as the heading of the work because that phrase was the named skeleton in the Master logs and the March 22 dumps. Almost every title followed the name instead of the request.

What was actually asked first was the study of root-level structures — how they arise, fail, and are maintained — and then of the neighboring families those roots live among: reality structures, structures people take to be real and literal, and simulation structures. CFTRL is one thing you intend to design using that study. It is not the territory being mapped.

If you do not have a reference frame for those structure families, CFTRL has nothing to fail inside of except a slogan. This file starts over at the correct layer: the structures, their frameworks and designs, what we presently believe about them, and how we will map them under incomplete information. CFTRL appears at the end as a consumer of the map, not as the name of the map.

2. How this topic arose

The Architect was working from the Master Change Log and Master Development Log. Many OAL entries have a name and a skeleton. Conditional Failure at the Root Level is one of those names. While thinking about that skeleton, the Architect saw that the supporting body was missing a prior layer: you cannot design failure-at-the-root well if you have not first studied roots, realities, literal-belief structures, and simulations as objects in their own right.

March 22, 2026 produced the first CFTRL dumps (OAL-CFTRL-NAIL-001; Development Review Dossier v1.0). September 23, 2026 produced a research request, an expansion about laboratories and incomplete maps, a request for an outline document, and then this correction: stop titling the study after the downstream framework.

3. Verbatim origin (Architect)

Correction statement, 23 September 2026, as received:

Everyone pretty much talked about conditional failure at the root level, which is fine, but what I actually asked for was the following: I noticed in every single title that it was almost with all of them conditional failure at the root level. The goal is to transition from thinking about conditional failure at the root level to:

- Root level structures
- Reality structures
- Structures that people think are real, literal
- Simulation structures
- Their framework, their design, things like that

The reasoning behind that is that if you do not have a map of reality as you know it—and this is not to say that I understand reality or that I know what reality or what being real is—but it is just to say that we have to have a reference frame. We have to be able to say we have studied this information. This is not an assertion that we understand in whole or in part that either we are in a simulation or we are not in a simulation. We understand what is real, what is not, what is base reality, what is a reality level structure. We are trying to learn more about root level structures, reality structures, and what we understand now so that we can use that information to help aid in the design and development of conditional failure at the root level. But we have to learn about root level structures, what we believe they are, what is considered real, what is considered not real, and what is considered a simulation. Can we distinguish what is reality from a simulation? How do we know? How do we build a map of reality? How do we build a map of a simulation? Do those architectures share the same structure? We have to ask all of these questions and then use the information towards what we are trying to design. That's just the basis that I have starting out.

The earlier prompts (study of roots; beyond-structure / laboratory / incomplete map; refusals; documentation for design) remain in force. They are now read as questions about these four structure families first. Full earlier verbatim text is in OAL-CFTRL-OUT-004 and the collaboration packet. This file does not re-title those questions as a CFTRL monograph.

4. Purpose and goal

4.1 Purpose

Build a reference frame. Record what this project currently means by root-level structures, reality structures, structures taken as literal, and simulation structures; how those families are designed and maintained; how they can be told apart in practice; how maps of them can be drawn without claiming completeness; and how that record later feeds CFTRL and related lifecycle design.

4.2 Goal

A working, versioned structure-map — not a finished ontology of existence — that the team can point to and say: we have studied this. Claims stay scoped to “what we understand now.” Distinguishing reality from simulation is treated as an open, graded question, not as a thesis to prove. CFTRL is listed among the uses of the map, not as the subject heading.

4.3 Refusals (unchanged)

- No assumption of a complete architecture of simulations, reality, or existence.
- No assumption of a unified definition of reality.
- No aim of total mapping or total comprehension.
- No project to compromise roots, simulations, or base reality.

- No how-to for those acts.

5. The four families (working distinctions)

These are working categories for the map. They can overlap in one object. Overlap is a finding, not a reason to collapse the list.

5.1 Root-level structures

A root-level structure is a structure whose continued integrity is prerequisite to the operation, interpretation, or validity of other structures inside a stated boundary. It is not “the oldest thing,” “the biggest thing,” or “whatever someone labeled root.”

How they arise (working):

- Compositional enclosure — slower, containing levels that constrain faster ones (cells in tissues; processes in an operating system).
- Allocative parenthood — a layer that budgets energy, compute, law, or attention for children (hypervisor, kernel, constitution, scheduler).
- Generative authorship — a layer that wrote the child’s rules (a simulation host; a founding protocol).
- Epistemic trust base — what a system must trust in order to decide what is true (clock, identity, log, attestation).
- Emergent enclosure — a structure that was not designed as a root and became one (a market, a language, a logistics standard).

How they are maintained: provisioning, rule enforcement, error correction, and some upward legibility. Starve the last two and dependents keep running on a lie.

How they fail: maintenance halt, allocation corruption, rule rewrite, identity or time failure, observation failure, visibility cut, scale snap, cascade across interdependent roots, recovery tools that live inside the failure, compromise (same uptime, changed objective). Failure of a root is not yet CFTRL. CFTRL is a later, narrower condition: root trouble plus dependents who cannot go up plus scale the architecture cannot absorb.

Root vs not-root (tests, not mood): substitution; blast radius; authority from below; visibility; allocation monopoly; error persistence across child resets; termination rights; interaction drop at the boundary. A bottleneck or a popular node is not a root if the dependents can rebuild its function from inside their own layer.

5.2 Reality structures

A reality structure, in this project, is a structure that belongs to the world-as-it-operates inside the reference frame we currently have — physical, biological, social, institutional — without a claim that the frame is complete or that “reality” has been defined for all speakers.

Examples of the sort of object, not a closed list: conservation and causal order as we model them; organisms and ecosystems; built infrastructure; legal and monetary systems as they actually constrain action; the fact that a door either opens or it does not regardless of the story told about it.

What this category is for: a place to put structures we treat as given for design purposes, while remaining free to later discover that a given item was a simulation structure or a literal-belief structure we had miscategorized.

What this category is not: a certificate that we know what being real is. The Architect’s sentence stands: this is not an assertion that we understand reality in whole or in part, nor that we are or are not in a simulation.

5.3 Structures people think are real and literal

A literal-belief structure is a structure that agents treat as the territory when it may only be a map, a story, a model, a brand, a metric, a sacred object, or an institutional performance. It can have real effects. Effects do not settle ontology.

This family exists because CFTRL, laboratories, and “maps of reality” will be distorted if the team cannot tell “this holds independently of our story about it” from “this holds because enough people act as if it does” from “this holds only inside a model.”

Working marks of a literal-belief structure:

- Its maintenance is mostly narrative, ritual, metric, or authority rather than substrate.
- Disagreement about it is treated as defect in the speaker, not as a possible defect in the object.
- It can be replaced by another story without a corresponding change in the underlying process — or it cannot, and that resistance is itself data.
- People use the word “literally” of it as a stop-sign rather than as a test.

Organizations, markets, scientific models, religions, national identities, dashboards, and “the root” itself can all spend time in this family. So can the sentence “we live in a simulation” and the sentence “we do not.” Both can be literal-belief structures while the underlying question remains open.

5.4 Simulation structures

A simulation structure is a structure whose states are generated according to rules hosted by another structure that retains allocation or termination rights over it — or that is built to stand in for such a process (a model, a digital twin, a nested virtual machine, an ancestor-simulation in the philosophical sense, a tabletop, a training environment).

Known simulations (E1): software we wrote, VMs we run, models we fit. Their outside is an engineering fact.

Candidate simulations (E3/E4): worlds for which a parent is hypothesized but not in our possession. The map must keep these off the “known simulation” page.

Design marks common to known simulation structures: explicit rule tables; schedulers; reset and snapshot; partial fidelity; an operator or at least a host process; a cost function someone chose. Live reality structures may lack some or all of those marks. That possible divergence is one of the questions this study exists to keep open.

6. Distinguishing the families

Ask	Reality structure (working)	Literal-belief structure	Known simulation structure
Who hosts it?	No operator we can point to inside the current frame.	A community, institution, or narrator.	A host process, team, or parent layer we can name.
Can it be reset?	Not by us, except as metaphor.	Yes, by changing the story or the metric.	Often yes: snapshot, rewind, respawn.
Where do rules come from?	Given, within the frame. Provenance not programmable by us.	Declared, taught, enforced socially.	Configured. There is a file, a protocol, a designer.
What does failure look like?	Invariant break in the world-as-modeled.	Loss of uptake; the story stops compelling action.	Host starvation, rule edit, crash, operator stop.
Can dependents rebuild it from below?	Usually no, if it is also a root.	Sometimes: a rival story.	No, not without the host — unless the “simulation” was only a name.

Table 1. Distinctions for mapping. A single object can sit in more than one column. Say so.

6.1 Can we distinguish reality from simulation? How do we know?

For known simulations: yes. We distinguish them because we host them, or because we have an independent channel onto the host (the operator's rack, the hypervisor dashboard). That is an engineering fact, not a cosmology.

For the world of the current frame itself: we cannot close the question from inside if a hypothetical parent can forge every observation. Glitches do not prove a parent. The absence of glitches does not prove base. A hypothesis that absorbs both is not a test.

What we can do instead, and what this study commits to doing:

1. Keep “known simulation,” “candidate simulation,” and “reality-structure-in-the-current-frame” on different pages of the map.
2. Pre-register what would count as a parent-signature and what would not.
3. Prefer rights and provenance over mood: termination rights, allocation rights, reset rights, configurable rules.
4. Allow the answer “undecided at this grade.” That is a valid map region.

So: we can distinguish a VM from a blade. We cannot, from inside a single world, finish the sentence “this cosmos is / is not a simulation” and call it a finding. The reference frame records that limit instead of filling it.

7. How to build the two maps

7.1 A map of reality (as we know it now)

Not a map of Everything. A versioned picture of the current frame.

- Boundary: what this version is a map of.
- Invariants we currently treat as given (and the instruments that watch them).
- Known roots inside that boundary, with the test that earned them the mark.
- Literal-belief structures in use, labeled as such so they are not smuggled into the invariant list.
- Unknown regions drawn as unknown.
- A disconfirmation list: what would force a revision.
- A date and a version number. Maps without versions become dogma.

Reliability here means: the map fails in public and remains usable for design. It does not mean correspondence to a finished cosmos.

7.2 A map of a simulation

For a known simulation the map can be more complete because the host is in scope:

- Host identity and rights (who may allocate, pause, reset, end).
- Rule table and what is explicitly not simulated.
- Fidelity claims and known shortcuts.
- Child roots inside the simulation (its own clocks, identities, allocators).
- What the child can observe about the host — usually little.
- Termination and snapshot policy.
- Cost and maintenance of the host.

A map of a candidate simulation (the cosmos-as-simulation hypothesis) cannot honestly copy this template. Doing so would invent a host. The candidate page lists hypothesized marks and the tests that have not been met.

7.3 Do those architectures share structure?

Working answer: they can rhyme. They must not be assumed identical.

Shared pattern (family resemblance): nesting; dependency; allocators; incomplete observability downward; failure that dependents misread as local; the possibility of more than one root (clock, power, identity, story).

Likely divergences: evolved reality structures often have emergent enclosure and no operator; known simulations have designers, cost functions, and resets; literal-belief structures can be replaced by speech in a way neither physics nor a running hypervisor can; physical law, if it is a reality-structure root, may have no maintenance window in the engineering sense.

Divergence protocol: for each comparison, list what transfers, what breaks, and what only works if you smuggle a designer into the analogue. If you have to smuggle the designer, you are no longer mapping that analogue. You are mapping the metaphor.

7.4 Basis of comparison under incomplete information

You compare maps to maps and maps to failure, not maps to The Real Map.

- Predictive surplus — which map forecasts the next break better?
- Independent instruments — a claim that survives two channels that do not share a scheduler.
- Substitution cost — if removing a candidate root changes nothing, it was not serving as root in this boundary.
- Repair asymmetry — local patches that will not hold raise the grade of “shared upstream.”
- Stability under new data.
- A written list of what would kill the current map.

Failure is the comparison engine that does not require omniscience. A structure that never fails in your records has either a coarse map or a short interval.

8. Laboratory (for structures, not for proving the cosmos)

Yes, a laboratory can be designed. Its job is to improve the maps and the lifecycle of structures we are allowed to touch. Its job is not to settle E4 cosmology.

- Zone S1 — Known-simulation workshop. Owned stacks only. Map the host, the child roots, observability, maintenance. Induce failure only here.
- Zone S2 — Reality-analogue library. Public historical cases (infrastructure cascades, institutional lockouts, model failures). Score rhyme vs divergence. Do not cause the cases.
- Zone S3 — Literal-belief clinic. Track structures the team or a subject population treats as literal; test whether they survive substitution of the story.
- Zone S4 — Residual / beyond. Version the current reality-map; log structured residuals; keep competing explanations alive.

Study of what lies beyond a structure means residue and silhouette: what the current map cannot account for. It does not mean entering a parent we do not own.

9. What works, what does not, at this layer

9.1 What works

- Four families kept distinct even when one object sits in two.
- “What we understand now” as the tense of every claim.
- Known simulation vs candidate simulation.
- Maps with holes and version numbers.
- Failure as a teacher of maps.
- CFTRL postponed until there is something for it to stand on.

9.2 What does not

- Titling the structure study “CFTRL” and hoping the structures appear in the footnotes.
- Using “real,” “literal,” “root,” and “base” as synonyms.
- Treating an unexplained event as proof of a parent world.
- Treating social uptake as ontology.
- Copying a known-simulation map template onto the cosmos and filling the host with speculation.
- A finished definition of reality as a prerequisite for work.

10. Questions this study has to keep asking

1. What, inside the current frame, do we mark as a root-level structure, and by which test?
2. What do we currently treat as a reality structure, and what would move it to another family?
3. Which structures in play are literal-belief structures, including our own project language?
4. Which simulations do we actually host, and which are only hypothesized?
5. What marks, if any, would distinguish a reality structure from a simulation structure when the host is not in our possession?
6. Do the architectures rhyme, and where does the rhyme fail?
7. What belongs on Reality-Map v0 vs Simulation-Map v0 vs the undecided page?
8. What structured residue would force a revision of either map?
9. Only after those: which findings change how we design conditional failure at a root?

11. How this later aids CFTRL (downstream only)

Once the maps exist, CFTRL can be designed against named objects instead of against a phrase.

- A root-failure model needs a list of actual root types, not one mythical Root.
- A detection rule needs to know whether the failing object is a reality structure, a simulation child, or a literal-belief structure whose “failure” is loss of uptake.
- A laboratory for CFTRL needs to know which failures may be induced (owned simulations) and which may only be studied after the fact (reality analogues).
- A Department of Endings needs to know which endings are model endings and which would be a how-to against a root we do not own.
- “Base reality compromised” cannot be an operational sentence until “base,” “reality,” and “compromise” have homes on the map.

That is the order the Architect specified: learn the structures and what we believe they are; then use that information toward what we are trying to design.

12. Assessment and next bodies

Where we are: the collaboration has a large CFTRL paper trail and a thin structure-map. This file is the first Grok document that titles and organizes the work as a structure / reference-frame study. Reality-Map v0 and Simulation-Map v0 do not yet exist as drawings. The four families are working distinctions, not stamped definitions.

Where we are going: versioned maps of the four families; a laboratory that studies structures; CFTRL as a later consumer.

Next documents, if asked:

1. Reality-Map v0 — one bounded picture with holes, invariants, and candidate roots.
2. Simulation-Map v0 — known simulations only, plus a separate undecided page.
3. Literal-Belief Register v0 — project language that may be treating maps as territory.
4. Then, and only as a use of those maps, CFTRL Working Spec.

13. Holding statement

Understood. The study is of root-level structures, reality structures, structures taken as real and literal, and simulation structures — their frameworks and designs — under a reference frame we admit is incomplete. We are not asserting that we know what being real is, or that we are or are not in a simulation. We are asserting that we will study what we now have, write it down with holes, and only then spend it on the design of conditional failure at a root. The previous titles put the cart on the cover. This file puts the structures on the cover.

Draft. Not canon. 23 September 2026.